MIL-W-21733A(AS)
24 January 1971
SUPERSEDING
MIL-W-21733(WP)
28 November 1958

MILITARY SPECIFICATION

WEBBING, TEXTILE, NYLON, PNEUMATIC LIFE PRESERVER

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the requirements for nylon textile webbing intended for use in the manufacture of various pneumatic life preservers.

1.2 Classification - The webbing covered by this specification shall be of the following types, as specified (see 6.2):

Type I. - Harness webbing
Type II. - Accessory webbing

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATION

Military

MIL-W-43334 Webbing and Tape, Textile, Packaging and Packing of

FSC 8305

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STANDARDS

Federal

FED-STD-191	Textile Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(When requesting specifications, standards, and publications, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

3.1 Materials -

* 3.1.1 Yarn - The nylon yarn shall be a bright, high tenacity, light and heat resistant polyamide prepared from hexamethylene diamine and adipic acid or its derivatives. It shall have a minimum melting point of not less than 244°C (471.2°F.) The yarn shall not be bleached in any manner or in any subsequent process.

3.1.2 Twist - The twist of the 840 denier single and final ply of 420 denier yarns shall be 2-1/2 to 3 turns per inch.

* 3.2 Physical requirements - The construction and physical properties of the finished webbing shall be as specified in Table I.

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TABLE I

Construction and Physical Requirements

	Type I	Type II
Width, inches	1-1/2 ± 1/16	1 ± 1/32
Weight, oz./linear yard, max.	1.07	0.40
Face and back warp yarns, min.	144	100
Binder ends, min.	34	-
Total ends, min.	178	100
Picks per inch, min.	48	40
Warp yarn, size and ply	840/1	420/1
Binder yarn, size and ply	840/1	-
Filling yarn, size and ply	840/1	840/1
Breaking strength, lbs, min.	1800	575
After 100 hours weatherometer exposure, percent loss, max.	15	15
Colorfastness to light	Good	Good

3.2.1 Weave - The type I webbing shall be double plain weave, one end as one, with a single filling. The binders shall be drawn to weave in groups of two. The sequence of warp ends and binders shall be 8 warp ends and 2 binders, one of the binder ends weaving 2 up and 2 down. The other binder end weaving 2 down and 2 up. The type II webbing weave shall be a 2 up and 2 down herringbone twill with one reversal at the center of the webbing.

* 3.3 Color - The color of the webbing shall be as specified by the procuring activity.

* 3.4 Nonfibrous materials - The water-soluble and chloroform-soluble materials shall be no more than 2.0 percent when tested as specified in 4.3.

* 3.5 pH - The pH value of the water extract of the dyed and finished webbing shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.3.

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* 3.6 Residual shrinkage - The residual shrinkage of the natural and dyed webbing shall be no more than 2.0 percent when tested as specified in 4.3.3.

3.7 Length and put-up - Unless otherwise specified each roll of webbing shall contain 90 to 110 yards. No roll shall contain more than 3 pieces and no piece shall be less than 10 yards in length.

* 3.8 Identification Tickets - Each roll of webbing shall have an identification ticket attached in accordance with MIL-W-43334.

* 3.9 Fiber identification - Each roll of webbing shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.10 Workmanship - The finished webbing shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to perscribed requirements.

* 4.1.1 Certificate of compliance - Where certificates of compliance are submitted in accordance with 4.3, they shall contain verifiable actual test and inspection data. The Government reserves the right to inspect and test the webbing to verify the validity of the certification.

* 4.2 Sampling and inspection for acceptance - Sampling and inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.2.1 Component and material inspection - In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

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* 4.2.2 End item examination - Defects found during the examination shall be classified in accordance with 4.2.2.1 through 4.2.2.3.

* 4.2.2.1 Yard-by-yard examination - The required yardage of each piece shall be inspected on both sides and visual defects classified as listed in table II. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the webbing, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof, in which it occurs. The sample unit for this examination shall be one linear yard. The sample size shall be in accordance with Inspection Level II of MIL-STD-105. The acceptable quality level shall be 2.5 for major defects and 10.0 for total defects (major and minor combined) per 100 units. The lot size shall be expressed in units of one linear yard each. An approximate equal number of yards shall be examined from each roll selected. The number of rolls from which the sample is to be selected shall be in accordance with table III of this specification.

* TABLE II

Classification of Defects

Defect		Major	Minor
Abrasion mark, bruise	Resulting in rupture of individual yarns or plies, distortion in the orientation of threads, dimensional distortion, areas visibly thinner than adjoining unaffected areas, or in nap sufficient to obscure the identity of the filaments in any yarn	X	
Broken or missing end, cut, hole, or tear	Any	X	
Broken or missing pick, cut, hole, or tear	Two or more within a linear inch, regardless of the size of missing portion	X	
	Single, missing for not more than 1/4-inch width		X
Coarse filling bar	Texture visibly lower than in adjoining unaffected area, or resulting in visibly increased stiffness or thickness of webbing 1/		X

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TABLE II (CONT'D)

	Defect	Major	Minor
Crease, edges folded or rolled	Webbing twisted or distorted, will not lay flat upon application of manual pressures		X
Drop-ply	Clearly visible, on more than 2 ends within same length and extending over 9 linear inches 1/	X	
	Clearly visible, on 1 or 2 ends within same length and extending over 9 linear inches 1/		X
Edges cut or frayed	Clearly visible 1/	X	
Edges beaded or corded	Visible increase in thickness or misformed selvage 1/	X	
Edges loopy	Forming clearly visible filling loops or edges tied loosely to body of webbing for 2 linear inches or more 1/	X	
Edges loose	Resulting in waviness, distortion in orientation of filling, or looseness along edges	X	
Edges tight	Resulting in visible tension along edge, or pucker, waviness, bagginess, or slackness that cannot be flattened by manual pressure 1/	X	
Fine or light filling bar	Clearly visible 1/		X
Float	Multiple, 1/2 inch or more in combined warp and filling directions; or single, floating over more than 1 inch	X	

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TABLE II (CONT'D)

	Defect	Major	Minor
Float (Cont'd)	Multiple, less than 1/2 inch in combined warp and filling direction; or single, floating over more than 1/2 inch but not more than 1 inch if in warp, over more than 1/4 inch of the width but not more than 1 inch if in filling		X
Heavy filling bar, heavy place	Visibly stiffer or thicker than adjoining unaffected webbing	X	
Hitch-back crack, open place	Clearly visible opening between adjoining picks or warpwise tension area over part of the width resulting in visibly light and heavy places 1/	X	
Jerked-in filling, slough off	Clearly visible		X
Kinks	More than 3 in any 9 linear inches, clearly visible on surface of webbing		X
Knots	Two or more knots protruding from the surface of the webbing	X	
	Untrimmed ends extending from surface of webbing or one knot		X
Mispick, skips (harness)	Resulting in widthwise repeated floats more than 1/4 inch long	X	
	Resulting in abrupt break in sequence of weave. Widthwise repeated floats 1/4 inch or less long		X
Mixed filling shade bar	Clearly visible		X

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TABLE II (CONT'D)

Defect		Major	Minor
Slack end	Two or more in same length, jerked-in between picks, or forming clearly visible loops on surface of webbing 1/	X	
	Single, jerked-in between picks, or forming clearly visible loops on surface of webbing 1/		X
Slub or slug, gout	More than twice the thickness of the yarn (or ply, if plied)		X
Spot, stain, streak (rust, dirt, oil, grease, dye)	Any uncleaned spot, stain, or streak clearly visible 1/		X
Smash	Any smash	X	
Tight end	Clearly visible 1/	X	
Tight pick or filling edge scalloped	Resulting in rolling of webbing; (any visible indentation of edge)	X	
Wavy or ridgy webbing	Clearly visible, will not flatten under manual pressure (resulting from uneven tension)	X	
Weak or tender spot	Any	X	
Wrong draw	Extending 9 linear inches or more	X	
	Extending less than 9 inches		X

1/ Clearly visible at normal inspection distance (approximately 3 feet).

4.2.2.2 Overall examination - Each defect listed below shall be counted no more than once in each roll examined. The sample unit for this examination shall be one roll. The sample size and acceptance number shall be as shown in table III.

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Defects

Uncleanliness throughout
Uneven weaving throughout
Shade (when other than natural)
  Off shade
  Poor penetration
  Mottled: Cloudy or streaky throughout
Not labeled in accordance with Textile Fiber Products Identification Act

TABLE III

Sample Size

Lot size in yards	Sample size in rolls	Maximum number of defects acceptable in sample
1200 or less 1/	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If a lot contains fewer than 3 rolls, each roll in the lot shall be examined.

4.2.2.3 Length examination -

4.2.2.3.1 Examination for length of individual roll - Each roll in the sample shall be examined for the defects listed below. The sample unit for this examination shall be one roll. The sample size and acceptance number shall be as shown in table III.

Defects

Gross length less than specified minimum length or more than specified maximum length
Gross length more than 2 yards less than gross length marked on roll label
Any piece less than 10 yards in length
Any roll containing more than 2 pieces

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4.2.2.3.2 Examination for total yardage in sample - The lot shall be unacceptable if the total of the actual gross length of rolls in the sample is less than the total of the gross lengths marked on labels.

4.2.3 Examination of preparation for delivery requirements - An examination shall be made in accordance with the provisions of MIL-W-43334 to determine that packaging, packing and marking comply with the section 5 requirements of this specification.

*4.3 Testing of the end item - The methods of tesing specified in FED-STD-191, wherever applicable and as listed in table IV shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. Except as otherwise specified, all tests shall be performed under standard conditions in accordance with FED-STD-191. All test reports shall contain the individual values utilized in expressing the final result. The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more units fail to meet any requirement specified. The lot size shall be expressed in units of 1 linear yard. The sample unit for testing shall be 20 yards of Type I webbing and 7 yards of Type II webbing.

TABLE IV

Test Methods

Characteristics	Test Method
Material identification	1530 1/
Melting point	1534 1/
Absence of bleaching	1/
Denier	4021 1/
Twist	4052 or 4054 1/
Width	5020
Colorfastness to light	5660
Weight	5041
Total ends	5050
Yarns per inch	5050
Ply	Visual 2/
Weave	Visual 2/
Nonfibrous material	2611
pH	2811
Breaking strength	4.3.1/5100
Breaking strength after weatherometer exposure	4.3.2
Residual shrinkage	4.3.3

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be accepted for the stated requirement.
2/ One determination per sample unit and the results reported as "pass" or "fail".

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4.3.1 Breaking strength - The machine shall be so operated that the pulling jaws or grips shall separate at the rate of 4 inches per minute under no load. A test shall consist of five specimens so broken, with no break of the five specimens being lower than 95 percent of the minimum breaking strength specified for that type of webbing. The breaking strength shall be reported as the average value so determined.

* 4.3.2 Resistance to weathering - Specimens of webbing shall be exposed in the accelerated weathering unit as specified in Method 5804 of FED-STD-191. A sufficient quantity of specimens to make five warp breakage tests shall be exposed. Corex D filters and Sunshine Carbons shall be used. The total exposure time shall be 100 hours. The spray heads shall be shut off during the entire exposure period. At the end of the exposure period the specimens shall be brought to equilibrium under standard conditions as defined in FED-STD-191. They shall then be tested for breaking strength as specified in 4.3.1. The percent of breaking strength loss shall be calculated as follows:

$$\frac{\text{Original breaking strength - breaking strength after aging}}{\text{Original breaking strength}} \times 100 = \text{percent of breaking strength loss}$$

* 4.3.3 Residual shrinkage - Three 24-inch full width specimens shall be selected from the sample unit. The specimens shall be brought to equilibrium under Standard Conditions as defined in FED-STD-191. Eighteen-inch gage marks shall then be placed on the specimens with an indelible marking medium. The specimens shall then be placed in a container of water maintained at 100° ± 2°F. and kept submerged at that temperature for 60-70 minutes. (NOTE: It may be necessary to attach weights to the specimens during the exposure period.) At the end of the exposure period, the specimens shall be removed and air dried without tension at a temperature not exceeding 105°F. After drying, the specimens shall be brought to equilibrium under Standard Conditions as defined in FED-STD-191, the distance between the gage marks remeasured and shrinkage calculated as follows:

$$\frac{\text{Original length - final length}}{\text{Original length}} \times 100 = \text{percent shrinkage}$$

5. PREPARATION FOR DELIVERY

5.1 Packaging - Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Levels A, B and C - Webbing, put up as specified, shall be packaged in accordance with the applicable requirements of MIL-W-43334.

5.2 Packing - Packing shall be level A, B or C as specified (see 6.2).

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5.2.1 Levels A, B and C - Webbing shall be packed in accordance with the applicable requirements of MIL-W-43334.

5.3 Marking - In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-W-43334.

6. NOTES

6.1 Intended use - The webbing is intended for use in the manufacture of various pneumatic life preservers.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Type of webbing required (see 1.2)

(c) Color required

(d) Length of roll required

(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2)

* 6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - AS
Project No. 8305-N083

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-W-21733A(AS) WEBBING, TEXTILE, NYLON, PNEUMATIC LIFE PRESERVER

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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